BEGINNERS



C language

100 programs

The best 100 C programs
for a first-year student

Tech_Talks

Why Learn the C Programming Language?

C is an excellent choice for getting introduced to the world of programming because it is simple and easy to learn. While some concepts may be challenging, overall, you won't face major difficulties when learning C. It familiarizes you with all the key programming concepts such as data types, variables, functions, arrays, strings, conditional statements, loops, input/output, data structures, and more, which are similar in other modern programming languages. If you are a beginner, you should be familiar with C/C++ if you want to succeed in interviews for internships or jobs. All service companies like TCS, Accenture, IBM, etc., recruit C developers.

1. **Highly Efficient**

The C programming language is popular for its efficiency and performance. Programs written in C run quickly and use system resources efficiently. C is used in embedded systems because in these systems, resources are limited, and performance is crucial. Some examples of embedded systems that you use in your daily life include TV remotes, game controllers, digital cameras, calculators, and appliances like refrigerators, washing machines, microwaves, etc. All these devices contain embedded systems.

2. System-Level Programming

C provides low-level control over hardware and memory, making it particularly suitable for system-level programming and the creation of operating systems.

3. Portable

C is also popular because of its portability. Code written in C can be compiled and executed on different platforms with few modifications required. So, if you want to run a program on various platforms, C should be your choice.

4. Easy to Learn

The learning curve for C is relatively short. The language has few keywords and concepts. The syntax is easy to memorize and use. The C compiler provides descriptive errors, making it easier to debug code. This is why beginners appreciate it.

5. Versatile

You can use C to create simple utility software as well as large-scale enterprise software.

Applications Built Using the C Programming Language

- Operating Systems: Many modern and popular operating systems are written using C, such as UNIX, Linux, and Windows, etc.

- Embedded Systems: As mentioned earlier, embedded systems have limited memory and require fast-executing code. Thus, C is the language of choice for them.

- Compilers and Interpreters: Programming languages like C++, Python, and Java have their compilers and interpreters written in C.

- Database Systems: MySQL and PostgreSQL have portions of their codebase written in C.

- Networking and Protocol Stacks: C is often used to build networking software. C is efficient and offers better control, which is why it is used to write TCP/IP stack software.

1. Swap of two variables :

```
#include <stdio.h>

int main() {
   int a = 5, b = 10, temp;
   temp = a;
   a = b;
   b = temp;

   printf("a = %d, b = %d", a, b);
   return 0;
}
```

2. Find the maximum of two numbers:

```
#include <stdio.h>

int main() {
   int a = 5, b = 10;
   if(a > b)
      printf("Maximum is %d", a);
   else
      printf("Maximum is %d", b);
   return 0;
}
```

3. Sum of the first N natural numbers:

```
#include <stdio.h>

int main() {
   int n = 10, sum = 0;
   for(int i = 1; i <= n; ++i) {
      sum += i;
   }
   printf("Sum = %d", sum);
   return 0;
}
```

4. Factorial of a number:

```
#include <stdio.h>

int main() {
   int n = 5;
   int fact = 1;
   for(int i = 1; i <= n; ++i) {
      fact *= i;
   }
   printf("Factorial = %d", fact);
   return 0;
}
```

5. Reverse of a number:

```
#include <stdio.h>

int main() {
   int n = 12345, reversed = 0;
   while(n != 0) {
      reversed = reversed * 10 + n % 10;
      n /= 10;
   }
   printf("Reversed number = %d", reversed);
   return 0;
}
```

6. Check if a number is prime:

```
#include <stdio.h>

int main() {
   int n = 29, flag = 0;
   for(int i = 2; i <= n/2; ++i) {
      if(n % i == 0) {
         flag = 1;
         break;
      }
   }
   if (flag == 0)
      printf("%d is a prime number.", n);
   else
      printf("%d is not a prime number.", n);
   return 0;
}
```

7. Find the greatest common divisor (GCD):

```
#include <stdio.h>

int main() {
   int n1 = 56, n2 = 98;
   while(n1 != n2) {
      if(n1 > n2)
         n1 -= n2;
      else
         n2 -= n1;
   }
   printf("GCD = %d", n1);
   return 0;
}
```

8. Check for palindrome:

```
#include <stdio.h>

int main() {
   int n = 121, original, reversed = 0, remainder;
   original = n;
   while (n != 0) {
      remainder = n % 10;
      reversed = reversed * 10 + remainder;
      n /= 10;
   }
   if (original == reversed)
      printf("%d is a palindrome.", original);
   else
      printf("%d is not a palindrome.", original);
   return 0;
}
```

9. Selection sort:

```
#include <stdio.h>

int main() {
   int array[5] = {64, 25, 12, 22, 11};
   int n = 5;
   for (int i = 0; i < n-1; i++) {
      int min_idx = i;
      for (int j = i+1; j < n; j++)
         if (array[j] < array[min_idx])
            min_idx = j;

      int temp = array[min_idx];
      array[min_idx] = array[i];
      array[i] = temp;
   }

   printf("Sorted array: \n");
   for (int i = 0; i < n; i++)
      printf("%d ", array[i]);
   return 0;
}
```

10. Simple calculator:

```
#include <stdio.h>

int main() {
   int a, b;

   // Entrée des deux nombres
   printf("Enter the first number: ");
   scanf("%d", &a);

   printf("Enter the second number: ");
   scanf("%d", &b);

   // Calcul et affichage des résultats
   printf("Sum: %d\n", a + b);
   printf("Difference: %d\n", a - b);
   printf("Product: %d\n", a * b);
   printf("Quotient: %.2f\n", (float)a / b);

   return 0;
}
```

11. Print the first N Fibonacci numbers:

```
#include <stdio.h>

int main() {
   int n = 10, t1 = 0, t2 = 1, nextTerm;
   printf("Fibonacci Series: %d, %d, ", t1, t2);
   for (int i = 1; i <= n; ++i) {
      nextTerm = t1 + t2;
      printf("%d, ", nextTerm);
      t1 = t2;
      t2 = nextTerm;
   }
   return 0;
}
```

12. Calculate simple interest:

```
#include <stdio.h>

int main() {
   float p, r, t, interest;

   // Entrée des données
   printf("Enter the principal amount: ");
   scanf("%f", &p);

   printf("Enter the rate of interest: ");
   scanf("%f", &r);

   printf("Enter the time period: ");
   scanf("%f", &t);

   // Calcul de l'intérêt simple
   interest = (p * r * t) / 100;

   // Affichage du résultat
   printf("Simple Interest: %.2f\n", interest);

   return 0;
}
```

13. Check if a number is even or odd:

```
#include <stdio.h>

int main() {
   int num;

   // Entrée du nombre
   printf("Enter a number: ");
   scanf("%d", &num);

   // Vérification si le nombre est pair ou impair
   if (num % 2 == 0)
      printf("Even\n");
   else
      printf("Odd\n");

   return 0;
}
```

14. Calculate the area of a circle:

```
#include <stdio.h>
#define PI 3.14159265358979323846

int main() {
   float radius, area;

   // Entrée du rayon du cercle
   printf("Enter the radius of the circle: ");
   scanf("%f", &radius);

   // Calcul de la surface du cercle
   area = PI * radius * radius;

   // Affichage de la surface
   printf("Area: %.2f\n", area);

   return 0;
}
```

15. Fill an array with integers entered by the user:

```
#include <stdio.h>

int main() {
   int n, i;

   // Demande la taille du tableau
   printf("Enter the number of elements: ");
   scanf("%d", &n);

   // Déclare un tableau de taille n
   int array[n];

   // Remplissage du tableau
   for (i = 0; i < n; i++) {
      printf("Enter element %d: ", i + 1);
      scanf("%d", &array[i]);
   }

   // Affichage du contenu du tableau
   printf("The elements in the array are: ");
   for (i = 0; i < n; i++) {
      printf("%d ", array[i]);
   }
   printf("\n");

   return 0;
}
```

16. Simple file management:

```
#include <stdio.h>

int main() {
   FILE *file;

   // Ouverture du fichier en mode écriture ("w")
   file = fopen("output.txt", "w");

   // Vérification si le fichier s'ouvre
correctement
   if (file == NULL) {
      printf("Error opening file!\n");
      return 1;
   }

   // Écriture dans le fichier
   fprintf(file, "Hello, this is a sample text.");

   // Fermeture du fichier
   fclose(file);

   return 0;
}
```

17. Reverse a string:

```
#include <stdio.h>
#include <string.h>

int main() {
   char str[100], rev[100];
   int len, i;

   printf("Enter a string: ");
   gets(str);

   len = strlen(str);

   for (i = 0; i < len; i++) {
      rev[i] = str[len - i - 1];
   }
   rev[len] = '\0';

   printf("Reversed string: %s\n", rev);
```

18. Find the largest number in an array:

```
#include <stdio.h>

int main() {
   int n, i, max;

   printf("Enter the number of elements: ");
   scanf("%d", &n);

   int array[n];

   printf("Enter the elements: ");
   for (i = 0; i < n; i++) {
      scanf("%d", &array[i]);
   }

   max = array[0];
   for (i = 1; i < n; i++) {
      if (array[i] > max) {
```

```
    return 0;
}
```

19. Calculate the factorial of a number:

```
#include <stdio.h>

int main() {
    int n, i;
    unsigned long long fact = 1;

    printf("Enter a number: ");
    scanf("%d", &n);

    for (i = 1; i <= n; ++i) {
        fact *= i;
    }

    printf("Factorial of %d = %llu\n", n, fact);

    return 0;
}
```

20. Display the multiplication table of a number:

```
#include <stdio.h>

int main() {
    int num, i;

    printf("Enter a number: ");
    scanf("%d", &num);

    for (i = 1; i <= 10; ++i) {
        printf("%d * %d = %d\n", num, i, num *
i);
    }

    return 0;
}
```

21. Convert a decimal number to binary:

```
#include <stdio.h>

int main() {
    int n, i;
    int binary[32];

    printf("Enter a decimal number: ");
    scanf("%d", &n);

    i = 0;
    while (n > 0) {
        binary[i] = n % 2;
        n = n / 2;
        i++;
    }

    printf("Binary: ");
    for (int j = i - 1; j >= 0; j--) {
        printf("%d", binary[j]);
    }
```

22. Calculate the power of a number:

```
#include <stdio.h>

int main() {
    int base, exp, result = 1;

    printf("Enter base: ");
    scanf("%d", &base);

    printf("Enter exponent: ");
    scanf("%d", &exp);

    for (int i = 1; i <= exp; i++) {
        result *= base;
    }

    printf("Result: %d\n", result);

    return 0; }
```

```
            max = array[i];
        }
    }

    printf("The largest element is: %d\n",
max);

    return 0;
}
```

```
    printf("\n");

    return 0;
}
```

23. Calculate the quotient and remainder of a division:

```
#include <stdio.h>

int main() {
    int dividend, divisor, quotient, remainder;

    printf("Enter dividend: ");
    scanf("%d", &dividend);

    printf("Enter divisor: ");
    scanf("%d", &divisor);

    quotient = dividend / divisor;
    remainder = dividend % divisor;

    printf("Quotient = %d\n", quotient);
    printf("Remainder = %d\n", remainder);

    return 0;
}
```

24. Display the numbers from 1 to 10:

```
#include <stdio.h>

int main() {
    int i;
    for (i = 1; i <= 10; i++) {
        printf("%d\n", i);
    }
    return 0;
}
```

25. Sum of the numbers from 1 to 10:

```
#include <stdio.h>

int main() {
    int i, somme = 0;
    for (i = 1; i <= 10; i++) {
        somme += i;
    }
    printf("La somme des nombres de 1 à 10
est %d\n", somme);
    return 0;
}
```

26. Display the even numbers from 1 to 20:

```
#include <stdio.h>

int main() {
    int i;
    for (i = 2; i <= 20; i += 2) {
        printf("%d\n", i);
    }
    return 0;
}
```

27. Multiplication table of a number:

```
#include <stdio.h>

int main() {
    int i, n;
    printf("Entrez un nombre pour afficher sa
table de multiplication: ");
    scanf("%d", &n);
    for (i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", n, i, n * i);
    }
    return 0;
```

28. Display a number triangle:

```
#include <stdio.h>

int main() {
    int i, j, n;
    printf("Entrez le nombre de lignes du
triangle: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) {
            printf("%d ", j);
        }
```

```
}
```

29. Display the numbers from 10 to 1 in descending order:

```
#include <stdio.h>

int main() {
   int i;
   for (i = 10; i >= 1; i--) {
      printf("%d\n", i);
   }
   return 0;
}
```

```
      printf("\n");
   }
   return 0;
}
```

30. **Display the day of the week (using switch-case) :**

```
#include <stdio.h>

int main() {
   int jour;
   printf("Entrez un numéro de jour (1 à
7) : ");
   scanf("%d", &jour);
   switch (jour) {
      case 1: printf("Lundi\n"); break;
      case 2: printf("Mardi\n"); break;
      case 3: printf("Mercredi\n"); break;
      case 4: printf("Jeudi\n"); break;
      case 5: printf("Vendredi\n"); break;
      case 6: printf("Samedi\n"); break;
      case 7: printf("Dimanche\n"); break;
      default: printf("Numéro de jour
invalide\n"); break;
   }
   return 0;
}
```

31. Convert a grade to a letter (using switch-case):

```
#include <stdio.h>

int main() {
   int note;
   printf("Entrez votre note (0 à 100) : ");
   scanf("%d", &note);
   switch (note / 10) {
      case 10:
      case 9: printf("A\n"); break;
      case 8: printf("B\n"); break;
      case 7: printf("C\n"); break;
      case 6: printf("D\n"); break;
      case 5:
      case 4: printf("E\n"); break;
      default: printf("F\n"); break;
   }
   return 0;
}
```

32. Simple calculator (using switch-case):

```
#include <stdio.h>

int main() {
   int choix;
   float a, b;
   printf("Choisissez une opération:\n");
   printf("1. Addition\n2.
Soustraction\n3. Multiplication\n4.
Division\n");
   scanf("%d", &choix);
   printf("Entrez deux nombres : ");
   scanf("%f %f", &a, &b);
   switch (choix) {
      case 1: printf("Résultat: %.2f\n", a +
b); break;
      case 2: printf("Résultat: %.2f\n", a -
b); break;
      case 3: printf("Résultat: %.2f\n", a *
b); break;
      case 4:
         if (b != 0) {
            printf("Résultat: %.2f\n", a / b);
```

33. **Display the months based on the number (using switch-case) :**

```
#include <stdio.h>

int main() {
   int mois;
   printf("Entrez un numéro de mois (1 à
12) : ");
   scanf("%d", &mois);
   switch (mois) {
      case 1: printf("Janvier\n"); break;
      case 2: printf("Février\n"); break;
      case 3: printf("Mars\n"); break;
      case 4: printf("Avril\n"); break;
      case 5: printf("Mai\n"); break;
      case 6: printf("Juin\n"); break;
      case 7: printf("Juillet\n"); break;
      case 8: printf("Août\n"); break;
      case 9: printf("Septembre\n");
break;
      case 10: printf("Octobre\n"); break;
      case 11: printf("Novembre\n");
break;
      case 12: printf("Décembre\n");
break;
      default: printf("Numéro de mois
invalide\n"); break;
   }
   return 0;
}
```

34. **Calculate the operation based on the option (using switch-case):**

```
#include <stdio.h>

int main() {
   int option;
   float nombre;
   printf("Choisissez une option:\n");
   printf("1. Carré\n2. Cube\n");
   scanf("%d", &option);
   printf("Entrez un nombre : ");
   scanf("%f", &nombre);
   switch (option) {
      case 1: printf("Carré: %.2f\n",
nombre * nombre); break;
      case 2: printf("Cube: %.2f\n",
nombre * nombre * nombre); break;
      default: printf("Option invalide\n");
break;
   }
   return 0;
}
```

35. **Display the type of triangle (using switch-case):**

```
#include <stdio.h>

int main() {
   int a, b, c;
   printf("Entrez les longueurs des trois
côtés du triangle : ");
   scanf("%d %d %d", &a, &b, &c);
   switch ((a == b) + (b == c) + (c == a)) {
      } else {
         printf("Erreur : Division par
zéro\n");
      }
      break;
   default: printf("Choix invalide\n");
break;
   }
   return 0;
}
```

36. **Calculate the fare based on age (using switch-case):**

```
#include <stdio.h>

int main() {
   int age;
   printf("Entrez votre âge : ");
   scanf("%d", &age);
   switch (age / 10) {
      case 0:
```

```
        case 3: printf("Triangle
équilatéral\n"); break;
        case 2: printf("Triangle isocèle\n");
break;
        case 1: printf("Triangle scalène\n");
break;
        default: printf("Ce n'est pas un
triangle\n"); break;
    }
    return 0;
}
```

37. Convert temperature units (using switch-case):

```
#include <stdio.h>

int main() {
    int choix;
    float temperature;
    printf("Choisissez une conversion:\n");
    printf("1. Celsius à Fahrenheit\n2.
Fahrenheit à Celsius\n");
    scanf("%d", &choix);
    printf("Entrez la température : ");
    scanf("%f", &temperature);
    switch (choix) {
        case 1: printf("Température en
Fahrenheit: %.2f\n", (temperature * 9 /
5) + 32); break;
        case 2: printf("Température en
Celsius: %.2f\n", (temperature - 32) * 5 /
9); break;
        default: printf("Choix invalide\n");
break;
    }
    return 0;
}
```

38. Display the season based on the month (using switch-case):

```
#include <stdio.h>

int main() {
    int mois;
    printf("Entrez le mois (1 à 12) : ");
    scanf("%d", &mois);
    switch (mois) {
        case 12:
        case 1:
        case 2: printf("Hiver\n"); break;
        case 3:
        case 4:
        case 5: printf("Printemps\n"); break;
        case 6:
        case 7:
        case 8: printf("Été\n"); break;
        case 9:
        case 10:
        case 11: printf("Automne\n");
break;
        default: printf("Mois invalide\n");
break;
    }
    return 0;
}
```

39. Display ASCII characters for codes 32 to 126:

```
#include <stdio.h>

int main() {
```

40. Display the ASCII code of a character entered by the user:

```
#include <stdio.h>

int main() {
```

```
    int i;
    for (i = 32; i <= 126; i++) {
        printf("Code ASCII %d : %c\n", i,
(char)i);
    }
    return 0;
}
```

41. **Display ASCII characters for codes 0 to 31 (control characters):

```
#include <stdio.h>

int main() {
    int i;
    for (i = 0; i <= 31; i++) {
        printf("Code ASCII %d : %c\n", i,
(char)i);
    }
    return 0;
}
```

42. Convert an ASCII code to a character:

```
#include <stdio.h>

int main() {
    int code;
    printf("Entrez un code ASCII (0 à 255)
: ");
    scanf("%d", &code);
    if (code >= 0 && code <= 255) {
        printf("Caractère correspondant au
code ASCII %d : %c\n", code,
(char)code);
    } else {
        printf("Code ASCII invalide\n");
    }
    return 0;
}
```

43. Display the ASCII codes for the letters A to Z:

```
#include <stdio.h>

int main() {
    char lettre;
    for (lettre = 'A'; lettre <= 'Z'; lettre++)
{
        printf("Code ASCII de '%c' : %d\n",
lettre, (int)lettre);
    }
    return 0;
}
```

44. Display the numbers from 1 to 10 (using do..while):

```
#include <stdio.h>

int main() {
    int i = 1;
    do {
        printf("%d\n", i);
        i++;
    } while (i <= 10);
    return 0;
}
```

45. Sum of the numbers from 1 to 10 (using do..while):

```
#include <stdio.h>

int main() {
    int i = 1, somme = 0;
    do {
        somme += i;
        i++;
    } while (i <= 10);
```

46. Display the even numbers from 2 to 20 (using do..while):

```
#include <stdio.h>

int main() {
    int i = 2;
    do {
        printf("%d\n", i);
        i += 2;
    } while (i <= 20);
```

```
    printf("La somme des nombres de 1 à
10 est %d\n", somme);
    return 0;
}
```

47. Calculate the factorial of a number (using do..while):

```
#include <stdio.h>

int main() {
    int n, i = 1, fact = 1;
    printf("Entrez un nombre : ");
    scanf("%d", &n);
    do {
        fact *= i;
        i++;
    } while (i <= n);
    printf("La factorielle de %d est %d\n",
n, fact);
    return 0;
}
```

49. **Multiply a number by 2 until the result is greater than 100:**

```
#include <stdio.h>

int main() {
    int nombre;
    printf("Entrez un nombre : ");
    scanf("%d", &nombre);
    do {
        nombre *= 2;
        printf("%d\n", nombre);
    } while (nombre <= 100);
    return 0;
}
```

48. Display the numbers from 10 to 1 in descending order (using do..while):

```
#include <stdio.h>

int main() {
    int i = 10;
    do {
        printf("%d\n", i);
        i--;
    } while (i >= 1);
    return 0;
}
```

50. Ask the user to guess a number (simple game):

```
#include <stdio.h>

int main() {
    int nombre, guess;
    printf("Devinez le nombre (entre 1 et
10) : ");
    scanf("%d", &nombre);
    do {
        printf("Essayez encore : ");
        scanf("%d", &guess);
        if (guess != nombre) {
            printf("Incorrect. ");
        }
    } while (guess != nombre);
    printf("Correct !\n");
    return 0;
}
```

51. Calculate the average of numbers entered until the user inputs a negative number:

```
#include <stdio.h>

int main() {
   int nombre, somme = 0, count = 0;
   do {
      printf("Entrez un nombre (négatif
pour terminer) : ");
      scanf("%d", &nombre);
      if (nombre >= 0) {
         somme += nombre;
         count++;
      }
   } while (nombre >= 0);
   if (count > 0) {
      printf("La moyenne est %.2f\n",
(float)somme / count);
   } else {
      printf("Aucun nombre valide
entré.\n");
   }
   return 0;
}
```

52. Find the divisors of a given number:

```
#include <stdio.h>

int main() {
   int nombre, i = 1;
   printf("Entrez un nombre : ");
   scanf("%d", &nombre);
   do {
      if (nombre % i == 0) {
         printf("%d est un diviseur de
%d\n", i, nombre);
      }
      i++;
   } while (i <= nombre);
   return 0;
}
```

53. Calculate the average of students' grades (with input):

```
#include <stdio.h>

int main() {
   int nombreEleves;
   float somme = 0.0, note, moyenne;

   // Demander le nombre d'élèves
   printf("Entrez le nombre d'élèves : ");
   scanf("%d", &nombreEleves);

   // Vérifier que le nombre d'élèves est
positif
   if (nombreEleves <= 0) {
      printf("Le nombre d'élèves doit
être positif.\n");
      return 1;
   }

   // Saisir les notes des élèves
   for (int i = 1; i <= nombreEleves; i++) {
      printf("Entrez la note de l'élève %d
: ", i);
```

54. Calculate the average of students' grades (without input):

```
#include <stdio.h>

int main() {
   // Définir les notes des élèves et le
nombre d'élèves
   float notes[] = {15.5, 17.0, 12.0, 19.5,
14.0};  // Liste des notes
   int nombreEleves = 5;  // Nombre
d'élèves (doit correspondre au nombre
de notes)

   float somme = 0.0, moyenne;

   // Calculer la somme des notes
   for (int i = 0; i < nombreEleves; i++) {
      somme += notes[i];
   }

   // Calculer la moyenne
   moyenne = somme / nombreEleves;

   // Afficher la moyenne
```

```
        scanf("%f", &note);

        // Ajouter la note à la somme
        somme += note;
    }

    // Calculer la moyenne
    moyenne = somme / nombreEleves;

    // Afficher la moyenne
    printf("La moyenne des notes est :
%.2f\n", moyenne);

    return 0;
}
```

55. Calculate the distance between two points (using the Pythagorean theorem):

```
#include <stdio.h>
#include <math.h>

int main() {
    float x1, y1, x2, y2, distance;
    printf("Entrez les coordonnées du
premier point (x1 y1) : ");
    scanf("%f %f", &x1, &y1);
    printf("Entrez les coordonnées du
second point (x2 y2) : ");
    scanf("%f %f", &x2, &y2);
    distance = sqrt((x2 - x1) * (x2 - x1) +
(y2 - y1) * (y2 - y1));
    printf("La distance entre les points est
: %.2f\n", distance);
    return 0;
}
```

56. Verify the Pythagorean theorem for a triangle:

```
#include <stdio.h>
#include <stdbool.h>

bool estTrianglePythagore(int a, int b,
int c) {
    return (a * a + b * b == c * c) || (a * a
+ c * c == b * b) || (b * b + c * c == a *
a);
}

int main() {
    int a, b, c;
    printf("Entrez les longueurs des côtés
du triangle (a b c) : ");
    scanf("%d %d %d", &a, &b, &c);
    if (estTrianglePythagore(a, b, c)) {
        printf("Le triangle est un triangle
rectangle.\n");
    } else {
        printf("Le triangle n'est pas un
triangle rectangle.\n");
    }
    return 0;
}
```

57. Calculate the sum of the squares of the first n natural numbers:

```
#include <stdio.h>

int main() {
   int n, somme = 0;
   printf("Entrez un entier positif n : ");
   scanf("%d", &n);
   for (int i = 1; i <= n; i++) {
      somme += i * i;
   }
   printf("La somme des carrés des
premiers %d entiers naturels est :
%d\n", n, somme);
   return 0;
}
```

58. Calculate the product of integers from 1 to n (factorial):

```
#include <stdio.h>

int main() {
   int n, produit = 1;
   printf("Entrez un entier positif n : ");
   scanf("%d", &n);
   for (int i = 1; i <= n; i++) {
      produit *= i;
   }
   printf("Le produit des entiers de 1 à
%d est : %d\n", n, produit);
   return 0;
}
```

59. Calculate the exponential of a number (Taylor series approximation):

```
#include <stdio.h>

int main() {
   int n, i;
   float x, fact = 1, exp = 1.0;
   printf("Entrez un nombre et le
nombre de termes pour l'approximation
: ");
   scanf("%f %d", &x, &n);
   for (i = 1; i <= n; i++) {
      fact *= i;
      exp += (x * i) / fact;
   }
   printf("L'approximation de
l'exponentielle de %.2f avec %d termes
est : %.5f\n", x, n, exp);
   return 0;
}
```

60. Calculate the least common multiple (LCM) of two numbers:

```
#include <stdio.h>

int pgcd(int a, int b) {
   while (b != 0) {
      int temp = b;
      b = a % b;
      a = temp;
   }
   return a;
}

int ppcm(int a, int b) {
   return (a * b) / pgcd(a, b);
}

int main() {
   int a, b;
   printf("Entrez deux entiers pour
calculer leur PPCM : ");
   scanf("%d %d", &a, &b);
   printf("Le PPCM de %d et %d est :
%d\n", a, b, ppcm(a, b));
   return 0;
}
```

61. Calculate the sum of the terms in an arithmetic progression:

```
#include <stdio.h>

int main() {
   int premierTerme, raison, n;
   printf("Entrez le premier terme, la
raison et le nombre de termes : ");
   scanf("%d %d %d", &premierTerme,
&raison, &n);
   int somme = n * (2 * premierTerme +
(n - 1) * raison) / 2;
   printf("La somme des %d termes de
la progression arithmétique est : %d\n",
n, somme);
   return 0;
}
```

62. Calculate the area of a triangle from its three sides (Heron's formula):

```
#include <stdio.h>
#include <math.h>

int main() {
   float a, b, c, s, aire;
   printf("Entrez les longueurs des trois
côtés du triangle : ");
   scanf("%f %f %f", &a, &b, &c);
   s = (a + b + c) / 2;
   aire = sqrt(s * (s - a) * (s - b) * (s - c));
   printf("L'aire du triangle est : %.2f\n",
aire);
   return 0;
}
```

63. Check if a number is a perfect square:

```
#include <stdio.h>
#include <math.h>

int main() {
   int nombre;
   printf("Entrez un nombre : ");
   scanf("%d", &nombre);
   int racine = (int)sqrt(nombre);
   if (racine * racine == nombre) {
      printf("%d est un carré parfait.\n",
nombre);
   } else {
      printf("%d n'est pas un carré
parfait.\n", nombre);
   }
   return 0;
}
```

64. Calculate the absolute value of a number:

```
#include <stdio.h>

int main() {
   int nombre;
   printf("Entrez un nombre : ");
   scanf("%d", &nombre);
   int absolue = (nombre < 0) ? -nombre
: nombre;
   printf("La valeur absolue de %d est :
%d\n", nombre, absolue);
   return 0;
}
```

65. Calculate the dot product of two vectors:

```
#include <stdio.h>

int main() {
   int x1, y1, x2, y2;
   printf("Entrez les coordonnées du
premier vecteur (x1 y1) : ");
   scanf("%d %d", &x1, &y1);
   printf("Entrez les coordonnées du
second vecteur (x2 y2) : ");
   scanf("%d %d", &x2, &y2);
```

66. Calculate the volume of a sphere:

```
#include <stdio.h>
#include <math.h>

int main() {
   float rayon, volume;
   printf("Entrez le rayon de la sphère :
");
   scanf("%f", &rayon);
   volume = (4.0 / 3.0) * M_PI *
pow(rayon, 3);
```

```
    int produitScalaire = x1 * x2 + y1 * y2;
    printf("Le produit scalaire des
vecteurs est : %d\n", produitScalaire);
    return 0;
}
```

67. Get and display the current date and time:

```
#include <stdio.h>
#include <time.h>

int main() {
    // Obtenir l'heure actuelle
    time_t maintenant;
    struct tm *infoTemps;

    // Obtenir l'heure actuelle
    time(&maintenant);
    infoTemps = localtime(&maintenant);

    // Afficher la date et l'heure actuelles
    printf("Date et heure actuelles :
%02d-%02d-%d %02d:%02d:%02d\n",
        infoTemps->tm_mday,    // Jour
du mois
        infoTemps->tm_mon + 1, // Mois
(de 1 à 12)
        infoTemps->tm_year + 1900, //
Année (de 1900 à l'année actuelle)
        infoTemps->tm_hour,    // Heure
(0 à 23)
        infoTemps->tm_min,     //
Minutes (0 à 59)
        infoTemps->tm_sec);    //
Secondes (0 à 59)

    return 0;
}
```

68. Find the roots of a quadratic equation:

```
#include <stdio.h>
#include <math.h>

int main() {
    float a, b, c, discriminant, racine1,
racine2;
    printf("Entrez les coefficients a, b et c
: ");
    scanf("%f %f %f", &a, &b, &c);

    discriminant = b * b - 4 * a * c;

    if (discriminant > 0) {
        racine1 = (-b + sqrt(discriminant)) /
(2 * a);
        racine2 = (-b - sqrt(discriminant)) /
(2 * a);
        printf("Les racines sont : %.2f et
%.2f\n", racine1, racine2);
    } else if (discriminant == 0) {
        racine1 = -b / (2 * a);
        printf("La racine est : %.2f\n",
racine1);
    } else {
        printf("Les racines sont
complexes.\n");
    }

    return 0;
}
```

69. Calculate nPr (permutations) :

```
#include <stdio.h>

int factorielle(int n) {
    int resultat = 1;
    for (int i = 1; i <= n; i++) {
        resultat *= i;
    }
    return resultat;
}

int main() {
```

70. Calculate nCr (combinations) :

```
#include <stdio.h>

int factorielle(int n) {
    int resultat = 1;
    for (int i = 1; i <= n; i++) {
        resultat *= i;
    }
    return resultat;
}

int main() {
```

```
    int n, r;
    printf("Entrez les valeurs de n et r : ");
    scanf("%d %d", &n, &r);
    int nPr = factorielle(n) / factorielle(n -
r);
    printf("nPr = %d\n", nPr);
    return 0;
}
```

71. Find the roots of a quadratic equation with coefficient validation:

```
#include <stdio.h>
#include <math.h>

int main() {
    float a, b, c, discriminant, racine1,
racine2;
    printf("Entrez les coefficients a, b et c
(a != 0) : ");
    scanf("%f %f %f", &a, &b, &c);

    if (a == 0) {
        printf("a ne peut pas être 0.\n");
        return 1;
    }

    discriminant = b * b - 4 * a * c;

    if (discriminant > 0) {
        racine1 = (-b + sqrt(discriminant)) /
(2 * a);
        racine2 = (-b - sqrt(discriminant)) /
(2 * a);
        printf("Les racines sont : %.2f et
%.2f\n", racine1, racine2);
    } else if (discriminant == 0) {
        racine1 = -b / (2 * a);
        printf("La racine est : %.2f\n",
racine1);
    } else {
        printf("Les racines sont
complexes.\n");
    }

    return 0;
}
```

```
    int n, r;
    printf("Entrez les valeurs de n et r : ");
    scanf("%d %d", &n, &r);
    int nCr = factorielle(n) / (factorielle(r)
* factorielle(n - r));
    printf("nCr = %d\n", nCr);
    return 0;
}
```

72. Identify a perfect square using a different method:

```
#include <stdio.h>

int main() {
    int nombre, i = 1;
    printf("Entrez un nombre : ");
    scanf("%d", &nombre);

    while (i * i < nombre) {
        i++;
    }

    if (i * i == nombre) {
        printf("%d est un carré parfait.\n",
nombre);
    } else {
        printf("%d n'est pas un carré
parfait.\n", nombre);
    }

    return 0;
}
```

73. Calculate nPr (permutations) using separate functions :

```
#include <stdio.h>

int factorielle(int n) {
   int resultat = 1;
   for (int i = 1; i <= n; i++) {
      resultat *= i;
   }
   return resultat;
}

int main() {
   int n, r;
   printf("Entrez les valeurs de n et r : ");
   scanf("%d %d", &n, &r);
   int nPr = factorielle(n) / factorielle(n -
r);
   printf("nPr = %d\n", nPr);
   return 0;
}
```

74. Calculate nCr (combinations) using separate functions :

```
#include <stdio.h>

int factorielle(int n) {
   int resultat = 1;
   for (int i = 1; i <= n; i++) {
      resultat *= i;
   }
   return resultat;
}

int main() {
   int n, r;
   printf("Entrez les valeurs de n et r : ");
   scanf("%d %d", &n, &r);
   int nCr = factorielle(n) / (factorielle(r)
* factorielle(n - r));
   printf("nCr = %d\n", nCr);
   return 0;
}
```

75. Find the roots of a quadratic equation with real numbers :

```
#include <stdio.h>
#include <math.h>

int main() {
   float a, b, c, discriminant, racine1,
racine2;
   printf("Entrez les coefficients a, b et c
: ");
   scanf("%f %f %f", &a, &b, &c);

   discriminant = b * b - 4 * a * c;

   if (discriminant > 0) {
      racine1 = (-b + sqrt(discriminant)) /
(2 * a);
      racine2 = (-b - sqrt(discriminant)) /
(2 * a);
      printf("Les racines réelles sont :
%.2f et %.2f\n", racine1, racine2);
   } else if (discriminant == 0) {
      racine1 = -b / (2 * a);
      printf("La racine réelle est :
%.2f\n", racine1);
   } else {
      printf("Pas de racines réelles.\n");
   }
```

76. Identify a perfect square using a function :

```
#include <stdio.h>
#include <math.h>

int estCarreParfait(int nombre) {
   int racine = (int)sqrt(nombre);
   return racine * racine == nombre;
}

int main() {
   int nombre;
   printf("Entrez un nombre : ");
   scanf("%d", &nombre);

   if (estCarreParfait(nombre)) {
      printf("%d est un carré parfait.\n",
nombre);
   } else {
      printf("%d n'est pas un carré
parfait.\n", nombre);
   }

   return 0;
}
```

```
    return 0;
}
```

77. Add two numbers (function) :

```
#include <stdio.h>

// Fonction pour ajouter deux nombres
int additionner(int a, int b) {
    return a + b;
}

int main() {
    int a, b;
    printf("Entrez deux nombres : ");
    scanf("%d %d", &a, &b);
    printf("La somme est : %d\n",
additionner(a, b));
    return 0;
}
```

78. Subtract two numbers (function) :

```
#include <stdio.h>

// Fonction pour soustraire deux
nombres
int soustraire(int a, int b) {
    return a - b;
}

int main() {
    int a, b;
    printf("Entrez deux nombres : ");
    scanf("%d %d", &a, &b);
    printf("La différence est : %d\n",
soustraire(a, b));
    return 0;
}
```

79. Multiply two numbers (function):

```
#include <stdio.h>

// Fonction pour multiplier deux
nombres
int multiplier(int a, int b) {
    return a * b;
}

int main() {
    int a, b;
    printf("Entrez deux nombres : ");
    scanf("%d %d", &a, &b);
    printf("Le produit est : %d\n",
multiplier(a, b));
    return 0;
}
```

80. Divide two numbers (function) :

```
#include <stdio.h>

// Fonction pour diviser deux nombres
float diviser(float a, float b) {
    if (b != 0) {
        return a / b;
    } else {
        printf("Erreur : Division par
zéro.\n");
        return 0;
    }
}

int main() {
    float a, b;
    printf("Entrez deux nombres : ");
    scanf("%f %f", &a, &b);
    printf("Le quotient est : %.2f\n",
diviser(a, b));
    return 0;
}
```

81. Calculate the square of a number (function) :

```
#include <stdio.h>

// Fonction pour calculer le carré d'un
nombre
```

82. Check if a number is even or odd (function) :

```
#include <stdio.h>

// Fonction pour vérifier si un nombre
est pair
```

```
int carre(int n) {
   return n * n;
}

int main() {
   int n;
   printf("Entrez un nombre : ");
   scanf("%d", &n);
   printf("Le carré de %d est : %d\n", n,
carre(n));
   return 0;
}
```

83. Calculate the factorial of a number (function) :

```
#include <stdio.h>

// Fonction pour calculer la factorielle
int factorielle(int n) {
   int resultat = 1;
   for (int i = 1; i <= n; i++) {
      resultat *= i;
   }
   return resultat;
}

int main() {
   int n;
   printf("Entrez un nombre : ");
   scanf("%d", &n);
   printf("La factorielle de %d est :
%d\n", n, factorielle(n));
   return 0;
}
```

84. Calculate the sum of the first n integers (function):

```
#include <stdio.h>

// Fonction pour calculer la somme des
n premiers entiers
int sommeNEntiers(int n) {
   int somme = 0;
   for (int i = 1; i <= n; i++) {
      somme += i;
   }
   return somme;
}

int main() {
   int n;
   printf("Entrez un nombre : ");
   scanf("%d", &n);
   printf("La somme des %d premiers
entiers est : %d\n", n,
sommeNEntiers(n));
   return 0;
}
```

85. Calculate the maximum of three numbers (function) :

```
#include <stdio.h>

// Fonction pour trouver le maximum
de trois nombres
int maximum(int a, int b, int c) {
   if (a >= b && a >= c) {
      return a;
   } else if (b >= a && b >= c) {
      return b;
```

86. Find the greatest common divisor (GCD) of two numbers (function) :

```
#include <stdio.h>

// Fonction pour calculer le PGCD
int pgcd(int a, int b) {
   while (b != 0) {
      int temp = b;
      b = a % b;
      a = temp;
   }
```

```
    } else {
        return c;
    }
}

int main() {
    int a, b, c;
    printf("Entrez trois nombres : ");
    scanf("%d %d %d", &a, &b, &c);
    printf("Le maximum est : %d\n",
maximum(a, b, c));
    return 0;
}
```

87. Check if a number is even or odd (function):

```
#include <stdio.h>

void verifierPairImpair(int n) {
    if (n % 2 == 0) {
        printf("%d est pair.\n", n);
    } else {
        printf("%d est impair.\n", n);
    }
}

int main() {
    int n;
    printf("Entrez un nombre : ");
    scanf("%d", &n);
    verifierPairImpair(n);
    return 0;
}
```

88. Display the address of a variable (pointer) :

```
#include <stdio.h>

int main() {
    int a = 10;
    int *ptr = &a;
    printf("Adresse de a : %p\n", ptr);
    return 0;
}
```

89. Modify the value of a variable using a pointer :

```
#include <stdio.h>

int main() {
    int a = 10;
    int *ptr = &a;
    *ptr = 20;  // Modifier la valeur de a
via le pointeur
    printf("Valeur de a : %d\n", a);
    return 0;
}
```

90. Pass an argument by pointer to a function:

```
#include <stdio.h>

void incrementer(int *p) {
    (*p)++;
}

int main() {
    int a = 5;
    incrementer(&a);
    printf("Valeur après incrémentation :
%d\n", a);
    return 0;
}
```

91. Swap the values of two variables using pointers :

```
#include <stdio.h>

void echanger(int *x, int *y) {
   int temp = *x;
   *x = *y;
   *y = temp;
}

int main() {
   int a = 5, b = 10;
   echanger(&a, &b);
   printf("a : %d, b : %d\n", a, b);
   return 0;
}
```

92. Use a pointer to access the elements of an array:

```
#include <stdio.h>

int main() {
   int tableau[3] = {1, 2, 3};
   int *ptr = tableau;
   for (int i = 0; i < 3; i++) {
      printf("Element %d : %d\n", i, *(ptr
+ i));
   }
   return 0;
}
```

93. Calculate the size of an array using pointers :

```
#include <stdio.h>

int main() {
   int tableau[5] = {1, 2, 3, 4, 5};
   int *ptr = tableau;
   int taille = (int)((char *)(&tableau[5]) -
(char *)ptr) / sizeof(int);
   printf("Taille du tableau : %d\n",
taille);
   return 0;
}
```

94. Dynamically allocate an array and fill it :

```
#include <stdio.h>
#include <stdlib.h>

int main() {
   int taille = 5;
   int *tableau = (int *)malloc(taille *
sizeof(int));
   if (tableau == NULL) {
      printf("Échec de l'allocation
mémoire.\n");
      return 1;
   }

   for (int i = 0; i < taille; i++) {
      tableau[i] = i + 1;
   }

   for (int i = 0; i < taille; i++) {
      printf("Element %d : %d\n", i,
tableau[i]);
   }

   free(tableau); // Libérer la mémoire
allouée
   return 0;
}
```

95. Use pointers in a function to modify multiple values:

```
#include <stdio.h>

void calculer(int *somme, int *produit,
int a, int b) {
   *somme = a + b;
   *produit = a * b;
}

int main() {
   int a = 5, b = 10;
   int somme, produit;
   calculer(&somme, &produit, a, b);
   printf("Somme : %d, Produit : %d\n",
somme, produit);
   return 0;
}
```

96. Use pointers in a function to return a string :

```
#include <stdio.h>

char *greeting() {
   char *message = "Bonjour, monde!";
   return message;
}

int main() {
   printf("%s\n", greeting());
   return 0;
}
```

97. Pointer to a pointer :

```
#include <stdio.h>

int main() {
   int a = 10;
   int *ptr1 = &a;
   int **ptr2 = &ptr1;
   printf("Valeur de a : %d\n", **ptr2);
   return 0;
}
```

98. Define and display information about a student (structure) :

```
#include <stdio.h>

// Définition de la structure pour un
étudiant
struct Etudiant {
   char nom[50];
   int age;
   float moyenne;
};

int main() {
   // Initialiser un étudiant
   struct Etudiant etudiant = {"Alice", 20,
15.5};

   // Afficher les informations de
l'étudiant
   printf("Nom : %s\n", etudiant.nom);
   printf("Âge : %d\n", etudiant.age);
   printf("Moyenne : %.2f\n",
etudiant.moyenne);

   return 0;
}
```

99. Calculate and display the area of a rectangle (structure) :

```
#include <stdio.h>

// Définition de la structure pour un rectangle
struct Rectangle {
   float largeur;
   float hauteur;
};

// Fonction pour calculer la surface du rectangle
float surface(struct Rectangle rect) {
   return rect.largeur * rect.hauteur;
}

int main() {
   // Initialiser un rectangle
   struct Rectangle rect = {5.0, 3.0};

   // Calculer et afficher la surface
   printf("Surface du rectangle : %.2f\n", surface(rect));

   return 0;
}
```

100. Store and display 2D point coordinates :

```
#include <stdio.h>

// Définition de la structure pour un point
struct Point {
   int x;
   int y;
};

// Fonction pour afficher les coordonnées d'un point
void afficherPoint(struct Point p) {
   printf("Coordonnées du point : (%d, %d)\n", p.x, p.y);
}

int main() {
   // Initialiser un point
   struct Point point = {10, 20};

   // Afficher les coordonnées du point
   afficherPoint(point);

   return 0;
}
```